

Question	Marking Guidance	Mark	Comments
08.1	To break down links between cells/cell walls OR To separate cells/cell walls OR To break down/hydrolyse cellulose/cell wall OR Allowing the stain to pass/diffuse into the cells OR Allowing the cells to be (more easily) squashed OR To stop mitosis;	1 (AO1)	Accept: To stop mitosis continuing Accept: To stop cell division Accept: To stop the cell cycle Ignore references to enzymes

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08.2	1. Eye protection; 2. Gloves; 3. Add <u>water</u> to spills (immediately); 4. Do not pour away down sink;	2 max (2 x AO1)	1. Accept wear goggles Ignore references to using dilute acid

Question	Marking Guidance	Mark	Comments
08.3	1. There is no difference between the mitotic index (at 200 and 300 μm); 2. t-test;	2 (2 x AO3)	1. Ignore reference to 'significant' 1. Reject correlation

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08.4	Correct answer within the range 19.95 to 20.03= 2 marks;; Incorrect answer of 57 or 57.2 (correct unit conversion and correct division) = 1 mark OR Incorrect answer of 2×10^{-5} (no unit conversion, correct division and correct $\times 0.35$) = 1 mark OR Incorrect answer of 2×10^x except 20 e.g.0.2/2/200/2000 (incorrect conversion, correct division and $\times 0.35$); = 1 mark	2 (2 x AO2)	

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08.5	Max 2 from 1. Roots/plant of the same age; 2. Same growing conditions (for all roots); 3. Same distance from root tip; 4. Same time in acid OR - Same temperature of acid; 5. Same concentration of acid; 6. Same pressure of squash; 7. Several fields of view (for each species); 8. Same time of day;	2 max (2 x AO3)	2. Accept description of growing conditions